

Itamar Avitan

PhD candidate · NeuroAI and computational cognitive neuroscience

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linkedin.com/in/itamar-avitan | scholar.google.com/citations?user=3fynQuAAAAAJ

RESEARCH INTERESTS

Representational alignment between deep neural networks and human vision; model recovery and identifiability in model comparison; evaluation metrics for model–brain and model–behaviour alignment.

EDUCATION

PhD candidate, direct track (integrated M.Sc.–PhD) · 2024–present

Ben-Gurion University of the Negev (BGU), Department of Industrial Engineering and Management; School of Brain Sciences and Cognition, Israel

- Advisor: Dr. Tal Golan, Brains and Machines Lab. Expected completion: 2029.
- Admitted to the direct PhD track for outstanding B.Sc. graduates; PhD candidacy examination passed, 2026.

M.Sc. in Industrial Engineering and Management, completed within the direct PhD track · 2026

Ben-Gurion University of the Negev, Israel

B.Sc. in Cognitive and Brain Sciences (computational track) and B.A. in Psychology · 2020–2024

Ben-Gurion University of the Negev, Israel

PEER-REVIEWED PUBLICATION

Itamar Avitan and Tal Golan (2025). Model–Behavior Alignment under Flexible Evaluation: When the Best-Fitting Model Isn't the Right One. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, main conference track.

Paper: arxiv.org/abs/2510.23321 | Code: github.com/brainsandmachines/oddoneout_model_recovery

Data: doi.org/10.6084/m9.figshare.30542690.v1 | Recorded presentation: slideslive.com/39047290

TALKS AND PRESENTATIONS

Sep 2026 · **Flash talk**, Neuro-AI-Talks (NEAT) 2026, Osnabrück, Germany. “System identification perspective on representational alignment” (with Maya Zach and Tal Golan).

Dec 2025 · **Poster**, NeurIPS 2025, San Diego, USA; poster session for the paper listed above.

Aug 2025 · **Contributed talk and poster** (with Tal Golan), Cognitive Computational Neuroscience (CCN) 2025, Amsterdam. “Rethinking Representational Alignment: Linear Probing Fails to Identify the Ground-Truth Model”; preliminary version of the NeurIPS 2025 paper. Session recording: youtube.com/watch?v=vT-3kV89Rhk&t=715s

RESEARCH EXPERIENCE

PhD researcher, Brains and Machines Lab · 2024–present

Ben-Gurion University of the Negev, Israel

- Built a large-scale model-recovery test for representational-alignment metrics: linearly aligned 20 diverse vision models to 4.5 million human odd-one-out judgments (THINGS dataset) and tested whether the data-generating model is re-identified from simulated responses.
- Showed that recovery accuracy plateaus below 80% even with millions of simulated trials; regression analyses linked misidentification primarily to shifts in representational geometry induced by the linear transformation. Flexible evaluation trades identifiability for predictive accuracy.

FELLOWSHIPS AND AWARDS

2024–present · **Lachish Fellowship for Outstanding Doctoral Students**, Kreitman School of Advanced Graduate Studies, BGU (five-year award).

2024–present · **Research Excellence Scholarship**, School of Brain Sciences and Cognition, BGU.

2025 · **Merit-based Travel Grant**, Data Science Research Center, BGU.

2025 · **Outstanding Poster Award**, Research Day, Department of Industrial Engineering and Management, BGU.

PROJECTS

Embodied Brain Technology Practicum (joint Brown University–BGU programme) · Jul–Aug 2026

Brown University, Providence, USA

- A two-week practicum for graduate students of both universities, entered by competitive selection, in which teams design, build and pitch a piece of neurotechnology. Our team of five built and pitched EarBetter, a biosignal-driven add-on for any headphones that filters the sounds which trigger anxiety.
- Code: github.com/Itamar-Avitan/earbetter-ebt-practicum

BCI4ALS: brain–computer interface (BCI) for a person with amyotrophic lateral sclerosis (ALS) · 2022–2023

BrainstormIL and IsrALS programme, Israel

- Team of five built an EEG-based P300 yes/no communication system, used live by the person it was built for: data collection, MNE-Python preprocessing, classifier search with scikit-learn, real-time prediction on OpenBCI hardware.
- Code: github.com/arbelna/BCI4ALS

TEACHING AND MENTORING

2023–present · **Teaching assistant**, BGU: Introduction to Cognition and Computation (2023/24, 2024/25, 2025/26); Deep Learning for Neuroscience and Cognition (spring 2026); Computational Approaches to Neuroimaging (2024).

2025–2026 · **Research mentoring**: two undergraduate research students in the lab (model–brain alignment; statistical modelling of neural spiking data).

2021–2023 · **Academic tutor**, Dean of Students Office, BGU: calculus, linear algebra and statistics.

SERVICE

2025 · Reviewer, Cognitive Computational Neuroscience (CCN) 2025, extended-abstracts track.

SKILLS AND LANGUAGES

Technical · Python, PyTorch, NumPy/SciPy/pandas, scikit-learn; Slurm cluster computing, Git, Linux.

Languages · Hebrew (native), English (fluent).

REFERENCES

Available on request.